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Values-led participatory design

Ole Sejer Iversen^{a*}, Kim Halskov^a and Tuck W. Leong^b

^aDigital Urban Living & CAVI, Aarhus University, Aarhus, Denmark; ^bCulture Lab, School of Computing Science, Newcastle University, Newcastle upon Tyne, UK

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The widespread use of participatory design (PD) has meant that different approaches and conceptualisations exist in this field today. In this article, it is argued that one fruitful approach is to rekindle a concern for values in design, focusing upon values as the engine that drives activities in PD. Drawing from the authors' own PD projects, this article shows how this can be accomplished: through designers enacting their appreciative judgement of values by engaging in a dynamic and dialogical process of cultivating the emergence of values, developing them and supporting their grounding.

Keywords: participatory design; human values; appreciative judgement of values

1. Engaging with values

When participatory design (PD) began nearly four decades ago amidst the explicitly political climate of Scandinavia, this design movement was forged with a commitment to values. PD 'makes explicit the critical, and inevitable, presence of values in the system development process' (Suchman 1993). And working with values in the design process is seen as an ethos that respects people's democratic rights (Ehn 1993), in that the people whose activity and experiences will ultimately be affected most directly by a design outcome ought to have a substantive say in what that outcome is. But as PD is embraced by many far beyond Scandinavia, infused with diverse traditions and disciplines (Kensing 2003, Muller 2009), this design practice has been transformed into one that is associated with a rich diversity of theories, practices, analyses and actions (Muller *et al.* 1997).

However, this transformation has generated some lively discussions and debates. One concerns the reification of methods and participation in PD practice, which may have shifted the focus away from values. Bødker and Iversen (2002) posit that some designers think that they are practising PD simply because they adopt particular participatory methods. But it is not only the use of participatory methods that makes a particular work PD. Instead, we will show in this paper, that it is about when, how and why these methods are used to engage with values that render the approach as being PD. Another confusion concerns the approach towards participation. While

*Corresponding author. Email: oiversen@cs.au.dk

participation is important in PD because it empowers stakeholders and allows them to feel connected to the design process, having stakeholders participate during the design process does not necessarily qualify it as PD. More importantly, PD is about negotiating values – a ‘moral proposition’ (Carroll and Rosson 2007) realised through participation.

In light of the above discussion, we want to present a different way to approach PD – an approach that cultivates values and bring values to the fore of the PD process. This means that values represent the engine that drives our approach to PD as well as being inherently embedded in every aspect of our work. This approach not only considers users’ and stakeholders’ values in the design process but also takes into account the values that designers bring to the PD process.

Such attention to values is not new. As we have indicated (and will extend later), many PD practitioners do engage with values. Values have also been proposed as a focal concern in human–computer interaction (Harper *et al.* 2008). This rekindling of values responds to its importance in co-operative design (Greenbaum and Kyng 1991). However, this paper presents a way to conceptualise this process and illustrates how we put this process to work. In doing so, we are suggesting not only that this values-led approach can address some of the misconceptions about PD but also that it is a fruitful path to pursue PD.

Embracing this values-led PD requires an important shift in how we behold methods and participation. While there is a need to wrestle with methods and participation in PD (Kensing 2003), reifying methods and participation can short-change what they can truly offer PD. For example, instead of worrying about which method to adopt in order to best capture stakeholders’ needs for design, or which methods can strengthen stakeholders’ ownership and commitment, the focus should be on how methods can best be used to work with values during the design process. In short, the shift we propose implores us to view methods and participation as means to achieving what we hold to be the ultimate ends of PD: a core engagement with values.

1.1. Aim and contribution

To illustrate how we work with values and demonstrate how methods and stakeholder participation can be used to engage with values, we will draw from three design cases: two in museums and one in a primary school. Our work with values is a dialogical process, where we orchestrate, facilitate and create opportunities for dialogue amongst stakeholders. It is through this dialogue that we cultivate the *emergence* of values, *develop* the values and *ground* the values during the design process.

Examples from these cases reveal how our concern for values influenced our choice of methods and in turn how we utilised the method to address values in design choices. This pervading concern for values also shaped the ways we worked with stakeholders, such as how we facilitated the negotiation of design dilemmas that arose from conflicting values. Further, we demonstrate how our concern for values transformed our design ideas throughout the design process as the values concerned are refined and eventually grounded in the final product.

For instance, the shared value of experiential and kinaesthetic learning in the primary school emerged in collaboration with designers through the use of various PD methods that offered stakeholders opportunities to dialogue under different

settings and in different contexts This value was further developed using PD methods that encouraged dialogue through collaborative games and problem-solving activities before it was eventually grounded through design.

Describing and explicating this dialogical process of emergence, development and grounding, this paper will contribute to furthering current PD practice. We also highlight the crucial role that the designer's *appreciative judgement system of values* plays in this process. Finally, providing further insights into this process will contribute to those wishing to engage with values in their PD practice. But first, we will clarify what we mean when we refer to values and position our approach to values in relation to other value-driven approaches, including its conceptualisation within the field of PD.

2. Values

2.1. Defining values

Values are enduring beliefs that we hold concerning *desirable modes of conduct* or *end-state of existence* in different situations, societies and cultural contexts (Rokeach 1973, Almond and Wilson 1988). A desirable mode of conduct could be taking care of loved ones, or being active and healthy; while a desirable end-state could be a preference for peaceful existence or democracy. Values have a transcendental quality, guiding and giving actions and decisions direction and imbuing them with emotional intensity (Schwartz 1994). They act as a goal and guide attitudes, judgements and comparisons across specific objects and situations and beyond immediate goals to more long-term goals (Rokeach 1973).

As such, our collection of values or value system is the generalised plan within whose frame or horizon we try to determine, from case to case, what is good or valuable, what is preferable and not, what we endorse or oppose, what we believe in and not. Not all judgements we make about our conduct are values. Our preferences for certain kinds of foods, for example, do not qualify as values since they do not implicate ultimate modes of behaviour or end-states of existence. Thus, values are fundamental to what makes us human (Rokeach 1973, Harper *et al.* 2008).

2.2. Values in participatory design

The Scandinavian tradition of PD is rooted in the Scandinavian trade union projects and the critical research tradition (Bansler 1989). It emerged as a critique of the negative impact of new technologies on people's working conditions as well as their health and safety at workplaces. The most prominent value underlying the Scandinavian tradition of PD was the ideal of workplace democracy (Bjerknes *et al.* 1987, p. 2, Bansler 1989, p. 14) with active participation of employees in order for them to have a real influence on their own working conditions. Closely tied to this were values of quality of working life and designing for skilled workers (Bjerknes *et al.* 1987, p. 2). When discussing a work-oriented design approach, Ehn (1988) saw the need to cultivate the value of emancipatory practice in PD. These early PD values from the 1980s had been consistently echoed in the early PDC conference, for example, by Suchman (Schuler and Namioka 1993).

Clearly, the value system of the Scandinavian tradition of PD is formed by a set of general, taken-for-granted and stable values shared by a community of researchers and design practitioners. Research and design practice were preoccupied with how

those values could be implemented in design processes. For instance, the tool perspective which originated from the Utopia projects emphasised designing for skilled work and gave them control of their works practice (Ehn 1988). Co-operative prototyping has been suggested as a design technique that enables users to actively participate and contribute in design processes (Bødker and Grønbæk 1991), and design collaboration has been proposed as a process model for enabling active user participation in design processes. Several frameworks have been suggested as a platform for gaining an overview of the methods and techniques at the disposal of PD practitioners (Kensing and Munk-Madsen 1993, Muller *et al.* 1993).

2.3. *Our approach to working with values*

While our approach is derived from this Scandinavian tradition of PD, an orientation that was also argued by Petersen *et al.* (2004), the values we work with are not limited to those discussed in our ‘historical’ account of PD. After all, those values were very much framed by the social and political forces of its time as well as its main concern with designing for the work setting. In contrast, values are emergent in our work, and as we will show later, these values can vary from one project to another. They are also not limited to predefined sets or categories.

So, the values we work with are not necessarily universal, nor do they adhere to the ‘twelve values of ethical import’ (Friedman and Kahn 2008). Neither do we refer to pre-existing categories such as ‘worthwhileness’ (Cockton 2008). Instead, we have an *a priori* commitment to cultivate the emergence and discovery of local expressions of values whilst being mindful of further expression of values during the course of the design process. This approach transcends the common mistake that associates PD with values such as participation, democracy and human welfare (Friedman and Kahn 2008). Thus, unlike other value-driven approaches (e.g. Cockton 2004, Friedman *et al.* 2006), the way we work with values means that they *emerge in collaboration* with stakeholders, with the values interacting recursively with the design process and permeating the entire process. The designer also brings values to the design process through ‘seeing as’, and through making design judgement, established through his or her design repertoire. This judgement is *during* and *after* the process of the design intervention, i.e. reflection in and on action (Schön 1983). However, in our cases, it is also *prior* to action, in our preparation of the design process.

In fact, our approach towards values resonates in many ways with Halloran and co-workers’ (2009) approach in their co-design work. However, we are much more explicit about the role of the designer and especially the designer’s appreciative judgement of values, which we find to pervade the entire process, from emergence to grounding. Our approach also further fleshes out how values are grounded in users’ practice and points to ways whereby we can support this grounding as part of a meaningful practice.

Nelson and Stolterman (2003) frame judgement as the key element in the process of design. By design judgement, they point to design competence as decision-making that is not dependent on rules of logic found within formal systems of inquiry. Rather, judgement is dependent on the accumulation of the experienced consequences of choices made in complex situations (Nelson and Stolterman 2003, p. 181). In this sense, design judgement resembles what Vickers (1995) denotes as appreciative judgement: the capacity to understand or appreciate a situation through

the discernment of what is to be considered as background and what is to be considered as foreground in the formation of a project. In this paper, we paraphrase Nelson and Stolterman (2003) and Vickers (1995), and address our concern for values in PD as a specific kind of design judgement, which we term *appreciative judgement of values*. We stress that this appreciative judgement of values is not a kind of black art, but rather that this judgement usually occurs in a dialogical process of emergence, development and grounding of values.

3. Three design cases

The three cases below represent PD projects that illustrate the different ways in which designers and stakeholders engaged with the values dialogically, and how this shaped the realisation of the final ‘product’. The values involved reflect the times when the project took place, and what was believed to be desirable and important by both the specific stakeholders and designers, while at the same time being grounded in and responsive to the individual project’s settings and context.

Each project was successfully completed, with the final products being research prototypes that are concerned with new technologies and new interaction styles that could support the use experience in various domains. One of them, the Wisdom Well, has grown from just a prototype to being a part of a school’s teaching practice. In all three, users were able to engage with the final product in playful and imaginative ways, providing meaningful alternatives to existing technologies in their respective domains.

3.1. Case 1: Interactive School Environment

This project took place between 2004 and 2006 and was funded by a national grant together with some private funds. The stakeholders were wide ranging. They included the local municipality, an interdisciplinary group of researchers (computer scientists, ethnographers, educational experts, etc.) and, from the school, teachers, management staff and students. Three commercial partners were also involved. The brief was to develop new IT-supported learning environment in primary schools. While we worked primarily with one school, teachers from other schools also participated.

We, the designers, developed the concept of the Wisdom Well. The Well is a 12 m² interactive floor on which the children practise their knowledge and skills in a playful and collaborative way. The computer application uses camera-tracking to locate children’s limb contact points, making the application controllable by body movement. The Wisdom Well was a result of ongoing work with kinaesthetic learning as a leading value. At the time of writing, the product is still in use and lighter versions have been taken up by other primary schools. See Grønbæk *et al.* (2007) for more details about the project (Figure 1).

3.2. Case 2: Children’s Literature Museum

In 2005–2006, we collaborated with a small low-hierarchy organisation (two full-time principal partners and a number of freelancers and subcontractors) called 7th Heaven. They specialise in producing exhibitions related to children’s literature. The brief was to design and produce two installations relating to the storytelling universe

of Norse mythology. This was to be a part of a centre for Scandinavian children's literature that opened in the city of Køge, Denmark, in early 2006.

The design we came up with was Balder's Funeral Pyre (Figure 2), an interactive corridor that children can traverse. On one side of the dark and narrow corridor is an immersive rear projection of fire that glows slowly at floor level. Sudden dramatic explosions of fire occur when someone walks into this corridor. See Halskov and Dalsgaard (2006) for further details about the project.

3.3. Case 3: Rune Stone Exhibition

From 2008 to 2009, we worked with Moesgaard Museum to explore how digital technology can support an installation regarding rune stones that is engaging to young people. The museum initiated the project and invited us to be a part of the project. The other stakeholders were museum curators, archaeologists and hardware vendors.



Figure 1. The Wisdom Well.



Figure 2. The installation of Balder's Funeral Pyre.

The design team developed an interactive installation – a RuneTable (Figure 3). The RuneTable installation allows visitors to create and decorate a digital rune stone. This can be placed in an interactive landscape alongside other rune stones that previous visitors have created. Visitors have to consider for themselves who they will erect the rune stone for and why. Moreover, they will choose where to place the stone in the interactive landscape. Thereby, the RuneTable facilitates interplay between visitors' everyday lives (personal relations, emotions, important personal experiences, etc.) and the heritage associations of rune stones. More details about the project can be found in Dindler and Iversen (2009).

4. Working with values: a dialogical process

Drawing from these cases, we will now discuss how we work with values during the design process: from cultivating the *emergence* of values, to supporting their *development* and *grounding* into current practice. These are not distinct phases with clearly defined beginnings and endings. Instead, the terms describe the predominant values-related activity that is facilitated by the designers at that time during the design process. Furthermore, although the terms describe the focus of the activities, the actual workings of each phase are not formulaic. On the contrary, we hope to reveal in our discussion that the uniqueness of each project requires constant diligence from designers who must respond with their appreciative judgement of values, such as the choice of particular methods and tools, or how to frame and facilitate the process. The crucial role of this judgement will be illustrated later.

The way we work with values is centred on dialogue. Jones *et al.* (2007) argue that dialogue is a crucial element in PD. Just like Jones *et al.*, we adopt dialogue in the way David Bohm applies it, whereby a person in dialogue 'may prefer a certain position but does not hold to it non-negotiably'. However, 'in ordinary discussions, people usually hold relatively fixed positions and argue in favor of their views as they try to convince others to change. At best this may produce agreement or



Figure 3. The RuneTable at Moesgaard Museum.

compromise, but it does not give rise to anything creative' (Nichol 2003, p. 295). So, a core task for designers is to facilitate and orchestrate this dialogue. And as we will illustrate later, values do not progress stepwise in one direction towards their grounding, but rather the emergence, development and grounding of values occur recursively over the course of the design process.

4.1. *Emergence of values*

In the Children's Literature Museum (CLM) project, educational values emerged early in the process from several meetings with the two principal partners of 7th Heaven. They visited our lab and we demonstrated previous projects including some outside the domain of museums. We also visited the stakeholders at their office, where we were shown some of their previous projects and illustrations from various children's books. During the meetings, everyone had the opportunity to express their opinions, preferences, particular commitments and orientations. Thus, values emerged through both explicit and tacit means: through dialogue, discussions, observations and our interpretations. These values became the basis for the design process.

For example, the stakeholders felt that it is important that children are not bombarded or spoon-fed with information but, instead, are given the opportunity and time to engage in reflection and interpretation of the exhibit for themselves. Concurrently, at this early stage, we were also aware of our own values. Our values partly reflected our research interest in designing for social interaction as well as physical interaction. Thus, our initial response to the design brief arose from the dialogue about these values; how these values could respond to the brief so that children visiting the centre could experience the settings and moods from Norse mythology rather than being told the specific myths and stories and, moreover, to ensure some aspect of 'slowness' so as to encourage time for reflection. The values that emerged were non-didactic, exploratory and based on experiential learning through reflection and introspection.

Unlike the museum project (CLM), the Interactive School Environment (ISE) project involved a large number of stakeholders. After meeting with each group separately, we found that they came with disparate viewpoints, interests and underlying values. So, we decided that the best way to cultivate the values was by hosting very large-scale workshops that brought everyone together. The workshops were highly facilitated and emphasised an inclusive atmosphere that promoted free participation. Various methods such as mock-ups, scenarios and sketching were used in shared tasks. Together, they allowed for stakeholders' values to emerge through their articulation of opinions, views, concerns, desires and so on. The emergent values then set a trajectory for the design process. However, stakeholders were not just passive participants during this process. They came to the process prepared and were actively engaged in shaping the design process. This can influence the emergence of values through the agenda that they bring to the design process together with their assumptions of how their personal and professional world can be changed.

During an initial workshop, a representative from each group gave a short talk about their view of IT-supported education. Over 300 pupils from the school also put together a large dossier that addressed their vision of a Classroom of the Future. Each stakeholder was then given an opportunity to express their visions and hopes

about technology, education and school (both as an institution and as a physical building) via a multi-faceted task that we facilitated. This was then presented to everyone within the workshop. What emerged was a shared articulation of values such as the importance of the human body in education, and an open, democratic and inclusive school environment based on shared resources. As a result, the design trajectory shifted from traditional classroom-based fixed technologies such as the monitor and mouse (with a limited use of the human body) towards pervasive technologies that provide means for full-body interactions. Whilst the trajectory was shaped by these emergent values, these values were in turn continually (and iteratively) refined, re-conceptualised and renegotiated throughout the entire process.

We found that designers, the participation of stakeholders and methods make important contributions to cultivating the emergence of values. But the kind of values that emerge depends on how designers orchestrate the design process; for example, the designer's response to the design brief. This means that the particular set of values that emerges is a result of how the designer looks at stakeholders' practice and, in turn, how stakeholders respond to the way the designer looks upon their practice. Thus, values that emerge are not already fully formed, or exist *a priori*, ready for designers to collect. Values emerge from a dialogical process between stakeholders and designers. Furthermore, values are not always explicit and can emerge when users are challenged (Halloran *et al.* 2009) and, in this case, via dialogue. In short, the designer's appreciative judgement of values involves the designer being aware of his or her values, and knowing how to use particular tools to engage stakeholders in dialogue, so as to support the cultivation of emergent values.

Through the process of cultivating the emergence of values, we are able to establish the prerequisites of the design project and, in turn, its trajectory within the design process. On the other hand, during the development of values, the process reveals how the emergent values can play out and materialise in a possible future.

4.2. Development of values

After emergent values are identified, we select appropriate methods, tools, and artefacts that address the domain with a view to developing the values. All emergent values are brought to the fore in the design process even if particular values conflict and present dilemmas to the process. Thus, development of values can take two different directions. Either direction requires designers to decide how to facilitate and orchestrate this process with stakeholders.

First, if everyone (stakeholders and designers) is pretty much in agreement concerning what they believe are the emergent values, then the development process focuses upon refining, clarifying and honing the values. This would also allow a clearer translation of values from abstract formulations to more concrete design concepts. This was the case with the CLM project.

The emergent values were agreed upon by all parties and not contested. To develop the emergent values, we set up an inspiration card workshop (Halskov and Dalsgaard 2006). Through playing with cards that are printed with images from Norse mythology and novel technologies, participants are prompted to reflect and verbalise their views of how they could envision the future design. This technique also allowed participants to express aspects of their preferences pertaining to this design. During the workshop our partners spontaneously introduced a number of

sources of inspirations, which were found to align with the emergent basic values, such as 'slowness' in interaction and opportunities to learn through discovery and interpretation. Clearly, in this case, the new inspirations that emerged during the development phase affirmed stakeholders' shared values, demonstrating how the development process allowed for a more specific and concrete conceptualisation of values.

On the other hand, if dilemmas are encountered, a dialogical development process is initiated in order to overcome the dilemmas. In fact, we often find that such dilemmas can offer design opportunities that lead to creative leaps in the design process. Appropriate methods and processes can be used as a springboard to help stakeholders to reimagine and re-engage with their values. The idea is to create opportunities for them to question and to renegotiate their values, potentially unfinalising their original perceptions of their values. Sometimes, this could even lead to new conceptualisations of their values.

In the interactive school project (and in contrast to the children museum), the new trajectory guided by the emergent values led to a dilemma that questioned the traditional role of the teacher as information provider. The trajectory pointed to a constructivist mode of education; one where the student is at the centre of the teaching practice instead of the then-practiced teacher-led model. So what will become of teachers if their core role is not to teach and to impart knowledge?

In response, we facilitated a series of workshops to deal with this (highly sensitive) dilemma. The aim was to discover ways that can allow all the stakeholders to reframe the situation, to transcend the dilemma. Bohm reminds us that such dilemmas arise not only because people bring different values but also because of their assumptions. Then what is called for is to 'suspend those assumptions, so that you neither carry them out nor suppress them. You don't believe them, nor do you disbelieve them; you don't judge them as good or bad' (Nichol 2003, p. 314).

The method that we chose to encourage this kind of suspension was fictional inquiry (Dindler and Iversen 2007). By having all stakeholders act out their traditional role in a fictional space, they were able to temporarily suspend their viewpoints and reframe this dilemma without feeling threatened. They also suspended their assumptions and viewpoints, and through their imagination found a way to resolve the dilemma by seeing a new role of the teacher: a game master. This is a reconceptualisation of the value. In acting out as game masters during the workshop, teachers saw how they could still play an active and important role as teachers while putting the pupil at the centre of the learning process.

These examples show that the development of values relies heavily on the designer's appreciative judgement of other groups' values, i.e. ways to facilitate the process, on knowing how different methods or tools can support stakeholders' working through dilemmas as well as to refine and hone their values through dialogue and even their reconceptualisation. At the same, the success depends on the degree to which stakeholders are willing to engage and participate in the process. So, in this project, the willingness of stakeholders to pursue the dilemma through our workshops was crucial in the development and reconceptualisation of values. Thus, development involves taking emergent values into a particular design trajectory whereby those values are further explored in a dialogical process with the stakeholders. This development can involve processes that refine emergent values, or even allow for the emergence of new values as seen in the case of the CLM. Development may also require working with dilemmas, as seen in the case of ISE.

But the process of development does not only occur in the minds of stakeholders. Similar to Halloran *et al.* (2009), the relationship of values to design is found to be dynamic. Designers find ways to introduce the values being developed into tools such as scenarios and prototypes so as to ground this process of refinement and negotiation. As values become more refined and concrete, so are the scenarios or prototypes produced. Hence, there is a tight coupling between the process of refining values and the realisation of the values through these tools. Using these methods to realise values within PD has been well documented (Bødker and Christiansen 1997) but of course the use of scenarios and prototype in itself does not guarantee that values would be developed. With the realisation of values, there is an opportunity to ground the values into users' current practice.

4.3. Grounding of values

The process of development and grounding are like the two sides of a coin. However, grounding is not a given even if the values have undergone significant development and have even been reconceptualised. The importance of grounding during PD has been highlighted by Kensing (2003) and Clement and Van den Besselaar (1993). Arriving at a stage whereby stakeholders question their values and even resulting in their reconceptualising their original values during the design process is fine, but values are only grounded when stakeholders can negotiate this new-found conceptualisation successfully (and in equilibrium) within their everyday practice.

With the CLM, the smooth development process allowed us to refine the values. Simultaneously, this allowed us to hone the design over the course of the process, narrowing down our translation of how the values could be realised. In order to support the grounding of the values in the museum's current practice, we turned to a variety of PD techniques. Future scenarios, early sketches and inspirational card workshops (to combine new interaction styles and everyday situations from the museum) were utilised as ways of providing stakeholders with 'tools to think with'. These supported stakeholders in seeing how the prototype could work within the existing museum space. The final prototype was deployed in the CLM and was adapted and incorporated into the museum's activities. Thus, values were grounded, as the stakeholders were able to see its use within their existing work practice.

In the case of the school project, we progressed from the emergent values, nourishing and moulding them during the development process. Similarly to the CLM case, a range of PD techniques was utilised to support the grounding of values in the existing school practice. Thus, a more profound support of the grounding process was initiated in an effort to help the new-found values to gain a foothold within the school's practice. We included a large number of teachers because, according to them, the departure from traditional classroom teaching with an emphasis on the role of the body in pupils' learning constituted a significant change of values in terms of their current teaching practice. Three initiatives were taken to support the grounding process.

First, we held a workshop in which stakeholders were able to try existing kinaesthetic games, whereby they could explore the potentiality of their own body. Second, we initiated two public presentations at local schools in which the entire staff could learn more about kinaesthetic learning. Finally, we invited the Minister of Education as well as local and national television to participate in the first ISE prototyping session. The purpose of going public at this very early stage was to elicit

public debate on this new set of values. The public prototyping session was our attempt to reach a broader audience for our concern for kinaesthetic learning. We have no evidence that these initiatives (as a supplement to existing PD tools) led to a successful grounding of the values. However, the final design – the Wisdom Well – was well accepted and is now a part of the school's education practice.

In contrast, in the case of the RuneTable, the grounding of values did not occur even though the emergent values were rigorously developed with the stakeholders. During the design workshops, museum curators developed interactive museum installations that reflected their values for learning through active engagement – a constructivist approach to learning that relied on the younger audiences' ability to 'learn by doing' through active engagement of both body and intellect during the museum visit. While the curators were very engaged and appeared to be exploring new conceptualisation of their values during the staged design process, they abandoned these new-found values for their original values. This occurred just before the final design was decided upon. In this case, a higher degree of end-user participation would not have swayed the curators' willingness to negotiate their underlying values.

The stakeholders withdrew their commitment to these values at the final stage even though prototypes and scenarios that reflected these developed values were used to realise the values. The final installation did contain elements of active engagement, but it was limited to a very fixed set-up in which the museum guests were only able to enter words on a touch screen. The final set-up differed significantly from the values pursued during the design sessions. While we cannot know the full extent of their reasons, it does suggest that we need to support stakeholders at the point when values are to be grounded. This may necessitate the extension of the PD process beyond the finalisation of the design artefact to include the introduction of the artefact into the existing practice. In doing so, PD will be concerned with changed practices that reach beyond simply the design of the artefact.

In the above cases, the designer's appreciative judgement of values should also include the grounding of values into users' existing practice. In the case of the Wisdom Well, this was carried out through various dissemination activities, and with the CLM via finalisation of the prototype on site.

4.4. Realisation of values

Now that we have discussed the conceptual process of how we work with values, we will show what happens during this process in terms of the design artefact. Realisation is when we have a physical manifestation of our process of working with values, such as the finalised design prototype. Realisation and grounding are mutually dependent in the sense that realisation may initiate the process of grounding, and successful realisation requires the initiation of grounding. This physical artefact is in fact a distillation of all our previous refinement of values beginning from emergence, through development to grounding.

For the CLM, Balder's Funeral Pyre (Figure 2) was designed to create an evocative mood. The darkness (interactive corridor) with the projection of fire at floor level piqued children's interests and fired their imagination before they entered it. Their engagement was further heightened when they entered the corridor with the sudden dramatic fire explosions erupting on the wall. We did think of having text snippets about Balder's myth appear as part of the fire but we stuck to the basic idea

of communicating the mood rather than the specific story, which reflected the value of exploratory learning. So through this evocative design (and not via texts), we were able to realise the value that children should learn through opportunities to have time to reflect and to interpret instead of being bombarded with or force-fed information.

We wish to emphasise that this realisation was used to support the grounding of values. Prototyping with eight children, all around six years old, at the museum helped us to see how the installation could best fit into the museum's practice. It was also an opportunity for the stakeholders to be present and discuss the installation in action. For example, something emerged during the prototyping session. When returning to the beginning of the entrance after traversing the corridor, some children found that their silhouettes (caused by the projector onto the corridor) were visible to other kids who were still walking inside the corridor. This led to a lot of laughing and playing among the children. Although this could potentially inspire or lead to a design that pursued a conventional theme-park style interaction, we did not go down this route because our partners felt that this did not fit in with the museum (and its practice). We also felt that this would not uphold the value of allowing for time for reflection; slowing the pace of interaction.

With the ISE the values developed were concerned with kinaesthetic learning (as a supplement to conventional classroom teaching) and learner-centred experiential learning (as opposed to teacher-centred learning). Throughout the design process, different mock-ups and prototypes were utilised to pursue the changes that these values would cause within the existing school practice. One of these initial design ideas was the concept of an interactive floor in which children could use the full faculty of their body as cursors to engage in a playful learning environment controlled by a computer application (see Figure 1).

An application developed for the Wisdom Well was the Stepstone application (Iversen *et al.* 2007). Here, learning through the body was enabled whereby children were able to test their school knowledge by stepping on the correct answers to a question. Initially, teachers used a traditional PC interface to develop the questions to be used in the Stepstone application. However, with the value of learner-centredness in mind, we redesigned the Stepstone system to run in an Internet browser so that the children could develop Stepstone applications themselves. The system is running today with children from different schools able to develop and compete in a game of Stepstone consisting of different tasks such as spelling and grammar games on the Wisdom Wells. The questions are generated by teachers and by pupils themselves and they are available online. This is an instantiation of the stakeholders' values of open, democratic and inclusive education.

With the RuneTable (Figure 3), earlier discussion has highlighted how the values that were developed did not make it into the final prototype. Yet, a compromised version was built despite the fact that the process was detached from the final product. The product did bear some hints of the values developed. For example, visitors were still able to create inputs onto the rune stone and having visitors move the rune stone from the holder to the table engendered some token of physical movement.

5. Appreciative judgement of values

The way in which we engage and work with values can be described as a process that involves cultivating the emergence of values, developing values and the grounding of

values. The process is always fuelled by dialogue between designers and stakeholders, be it through formal meetings or more playful situations such as workshops. The designer's appreciative judgement of values facilitates this process, and determines when one phase ends, begins or repeats.

Our concern for values begins with our cultivating its emergence. This means finding ways to help participants to visualise and verbalise their ideas. Some of the ways we achieve this are through meetings, discussion, observations and workshops, as well as our interpretation of the situation, whether explicitly or tacitly. What stakeholders bring to the process, such as artefacts, ideas, previous inspiration or even previous work, can further influence the type of values that emerge. Similarly, designers come to the process with their own ideas, expectations and values. Their previous portfolio or even their design traditions could influence this. The key is that all emergent values are tabled and are opened up for dialogue.

Developing values basically involves the recursive process of refining the emergent values and translating them towards more concrete design ideas. Translation and refinement occur in tandem. The designer requires a nuanced understanding of his or her design repertoire to know which tools could be employed most effectively to approach the values. Again, the things that stakeholders bring, such as artefacts and ideas, can also influence the direction of this process. However, if the emergent values present a dilemma, then efforts will be channelled towards finding ways to transcend the dilemma. Methods whereby stakeholders can suspend their assumptions and judgement in a fun and imaginative way can energise creative reconceptualisations of the values that led to the dilemma in the first place. As described earlier, the reconceptualised values are also refined and, in the process, translated.

Grounding only occurs if the 'developed values' are perceived by the designer as being able to be comfortably integrated and exist in equilibrium with stakeholders' current practice. It could be the acceptance and use of a prototype in an existing practice. But developed values may not have the opportunity to be grounded if, for any reason, the stakeholders do not adopt these values in their practice. This was highlighted by the RuneTable case. In order to facilitate the grounding of values, PD practitioners could perhaps extend the design process beyond the finalisation of the artefact. This idea was hinted by Carroll *et al.* (2000), and also resonates with Christiansen's (1996) notion of the gardening attitude.

To reiterate, appreciative judgement of values requires introspection. By being introspective, the designer will become aware of her (or his) values, and how they colour, influence and guide her choice of the methods, resources and ways of working with stakeholders. Introspection also influences the ways in which she orchestrates the design process. This awareness will also help her to be more explicit about her design intuition and design judgements. Thus, her values will have a great impact on the design process. This includes considerations of how meetings are framed, facilitated and orchestrated. Furthermore, the choice of methods, including the props, artefacts used for this initial meeting and even the way that the invitation is framed must be carefully considered. This consideration is guided by the designers' response to the design brief and designers' understanding of users' current practice. Again, this is also coloured by the values that designers bring to the brief.

Designers need to be familiar with various methods and the ways they can be put to work with values. After all, the emergence of values, their development and grounding are highly sensitive to the type of methods used. So, there should always

be an alignment between the methods used and the designers' grounding perspective. As designers, we should ask ourselves why we use a particular method and how this method can best serve our work with values at each phase. In our cases, we found that stakeholders were more likely to be engaged if we employed methods that were often playful and appealed to their imagination.

6. Conclusion

We began this paper by describing some of the concerns pervading PD today: debates about methods and participation and under what circumstances their applications are considered to be core concerns to PD. In response, we argue that one fruitful approach is to use them reflectively to work with values in PD. While we acknowledge that methods to encourage participation in the design process are important, we argue for a different starting point. This is to focus instead upon values. Although some PD practitioners are already engaged with values as their core concern, we argue the need to engage with values more explicitly, to view values as the engine that drives our design efforts. That is why this article was given the title 'Values-led participatory design'. Values are, after all, fundamental to what makes us human.

But despite knowing this, values are often something that is taken for granted in the design process; something that is desirable but too hard to achieve in 'real' design activities. Furthermore, there appears to be a lack of understanding as to how we actually work with values. Thus, by drawing from our projects, we demonstrate how this might look like in practice. This includes cultivating, nourishing and moulding values through the iterative and dialogical process of emergence, development and grounding. This process is one way to conceptualise how we can work with values in PD. The way for designers to engage with this dialogical process is through harnessing their appreciative judgement of values, for it is only through exercising this judgement that methods and tools used in participative design processes take on particular significance.

We do not claim that this appreciative judgement of values is a new idea. What we did was to create this term from 'appreciative system', a term originally coined by Vickers (1995). When we use the phrase 'appreciative judgement of values', we wish to emphasise the importance of values and how this appreciative system can provide designers with a sensibility to engage with values in design.

By presenting a detailed discussion of how we work with values, we have laid out an approach that is different from other value-centred and value-sensitive approaches to designing technologies such as those proposed by Cockton (2004) and Friedman *et al.* (2006). Our work treats values as emergent and dynamic, and we engage with them during design through a dialogical process. This approach is similar to the way in which Halloran *et al.* (2009) work with values in co-design. However, as our paper illustrates, we are not only presenting a unified framework of seeing the process, but also want to highlight the crucial role of the designer; not only the values she (or he) brings to the process but in particular how her appreciative judgement of values colours and pervades her entire stance in the design life cycle. Furthermore, this paper takes an additional interest in explicating how we can support the grounding of values during this process. However, values are inherently vague and difficult to isolate. Thus, future work to better explicate human values will certainly strengthen the approach that we have proposed here. Work towards more

detailed descriptions – accounts and examples of processes of how we can engage with values-led design – could also deepen our understanding and further this agenda. Currently, PD practitioners are absent when stakeholders are trying to negotiate their new-found values, when they are trying to include the design artefact into their current practice. So, one future direction for PD as a field may be to pursue better understanding of how to support this part of the process.

We see this paper as a first step in describing a more robust approach towards values-led PD. This paper contributes towards illustrating an approach to work with values as the organising principle for PD. More importantly, this approach illustrates how we can work productively to support the dynamism of values during the design process. As such, we believe that this approach can also contribute to those who work with values in design beyond the PD tradition, such as those involved in co-design activities.

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References

- Almond, B. and Wilson, B., eds., 1988. *Values: a symposium*. Atlantic Highlands, NJ: Humanities Press International.
- Bansler, J., 1989. Systems development in Scandinavia: three theoretical schools. *Scandinavian Journal of Information Systems*, 1, 3–20.
- Bjerknes, G., Ehn, P., and Kyng, M., eds., 1987. *Computers and democracy – A Scandinavian challenge*. Aldershot, UK: Avebury.
- Bødker, S. and Christiansen, E., 1997. Scenarios as springboards in design. In: G. Bowker et al., eds., *Social science research, technical systems and cooperative work*. Mahwah, NJ: Erlbaum, 217–234.
- Bødker, S. and Grønbæk, K., 1991. Design in action: from prototyping by demonstration to cooperative prototyping. In: J. Greenbaum and M. Kyng, eds. *Design at work: cooperative design of computer systems*. Hillsdale, NJ: Lawrence Erlbaum Associates, 197–218.
- Bødker, S. and Iversen, O.S., 2002. *Staging a professional participatory design practice*. In: *Proceedings of NordiCHI*. Aarhus, Denmark. New York: ACM Press, 11–18.
- Carroll, J. and Rosson, M.B., 2007. Participatory design in community informatics. *Design Studies*, 28, 243–261.
- Carroll, J.M., et al., 2000. The development of cooperation: five years of participatory design in the virtual school. In: *Designing interactive systems*. New York: ACM Press, 239–251.
- Christiansen, E., 1996. *A gardening attitude. A case study of technical support work*. Menlo Park, CA: Institute for Research on Learning.
- Clement, A. and Van den Besselaar, P., 1993. A retrospective look at PD projects. *Communications of the ACM*, 36 (6), 29–37.
- Cockton, G., 2004. Value-centred HCI. In: *Proceedings of NordiCHI*. Tampere, Finland. New York: ACM Press, 149–160.
- Cockton, G., 2008. Designing worth – connecting preferred means to desired ends. *Interactions*, 15 (4), 54–57.
- Dindler, C. and Iversen, O.S., 2007. Fictional inquiry–design collaboration in a shared narrative space. *Co Design*, 3 (4), 213–234.
- Dindler, C. and Iversen, O., 2009. Motivation in the museum – mediating between everyday engagement and cultural heritage. In: *Proceedings of Nordes*. Oslo: Nordic Design Research, 1–10.

- Ehn, P., 1988. *Work-oriented design of computer artifacts*. Stockholm: Arbetslivscentrum.
- Ehn, P., 1993. Scandinavian design: on participation and skill. In: D. Schuler and A. Namioka, eds. *Participatory design: principles and practices*. Hillsdale, NJ: Lawrence Erlbaum Associates, 44–77.
- Friedman, B. and Kahn, P., Jr, 2008. Human values, ethics and design. In: A. Sears and J. Jacko, eds. *The human–computer interaction handbook: fundamentals, evolving technologies*. New York: Lawrence Erlbaum Associates, 1241–1266.
- Friedman, B., Peter Kahn, J., and Borning, A., 2006. Value sensitive design and information systems. In: P. Zhang and D. Galletta, eds. *HCI in management information systems: foundations*. New York: M.E. Sharpe, 348–372.
- Greenbaum, J. and Kyng, M., eds., 1991. *Design at work: cooperative design of computer systems*. Hillsdale, NJ: Lawrence Erlbaum Associates.
- Grønbæk, K., et al., 2007. IFloorGame: a platform for co-located collaborative games. In: *International conference on advances in computer entertainment technology*. Salzburg, Austria. New York: ACM Press, 64–71.
- Halloran, J., Hornecker, E., and Stringer, M., 2009. The value of values: resourcing co-design of ubiquitous computing. *CoDesign*, 4 (5), 245–273.
- Halskov, K. and Dalsgaard, P., 2006. Inspiration card workshops. In: *6th conference on designing interactive systems*. University Park, PA: ACM Press, 2–11.
- Harper, R., et al., eds., 2008. *Being human: Human–computer interaction in the year 2020*. Cambridge, UK: Microsoft Research.
- Iversen, O.S., et al., 2007. Stepstone: an interactive floor application for hearing impaired children with a cochlear implant. In: *6th international conference on interaction design and children*. Aalborg, Denmark. New York: ACM Press, 117–124.
- Jones, P., Christakis, A., and Flanagan, T., 2007. Dialogic design for the intelligent enterprise: collaborative strategy, process, and action. In: *International symposium of the international council on systems engineering*. INCOSE. San Diego, CA, 1–16.
- Kensing, F., 2003. *Methods and practices in participatory design*. Copenhagen: ITU Press.
- Kensing, F. and Munk-Madsen, A., 1993. PD: structure in the toolbox. *Communications of the ACM*, 36 (6), 78–85.
- Muller, M., 2009. Participatory design: the third space. In: A. Sears and J. Jacko, eds. *Human–computer interaction: development process*. Boca Raton, FL: CRC Press (Taylor and Francis).
- Muller, M., Wildman, D., and White, E., 1993. Taxonomy of PD practices: a brief practitioner’s guide. *Communications of the ACM*, 36 (4), 25–28.
- Muller, M., Haslwanter, J.H., and Dayton, T., 1997. Participatory practices in the software lifecycle. In: M. Helander and P. Landauer, eds. *Handbook of human–computer interaction*. Vol. 2. Amsterdam: Elsevier Science.
- Nelson, H.G. and Stolterman, E., 2003. *The design way*. Englewood Cliffs, NJ: Educational Technology Publications.
- Nichol, L., ed., 2003. *The essential David Bohm*. London: Routledge.
- Petersen, M., et al., 2004. Embracing values in designing domestic technologies. In: *Proceedings of European conference on cognitive ergonomics*. York, UK. New York: ACM Press, 161–171.
- Rokeach, M., 1973. *The nature of human values*. New York: Free Press.
- Schön, D.A., 1983. *The reflective practitioner*. New York: Basic Books.
- Schuler, D. and Namioka, A., eds., 1993. *Participatory design: principles and practices*. Hillsdale, NJ: Lawrence Erlbaum Associates.
- Schwartz, S., 1994. Are there universal aspects in the structure and contents of human values? *Journal of Social Issues*, 50 (4), 19–45.
- Suchman, L., 1993. Foreword. In: D. Schuler and A. Namioka, eds. *Participatory design: principles and practices*. Hillsdale, NJ: Lawrence Erlbaum Associates, vii–ix.
- Vickers, G., 1995. *The art of judgement: a study of policy making*. Thousand Oaks, CA: Sage.